

EE2080: Random Processes

Practice Problem Set 1

August 13, 2025

1. Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $Y_1, Y_2, Y_3, \dots$ be a sequence of i.i.d. random variables defined w.r.t $\mathcal{F}$ with $\mathbb{E}[Y_i] = 0$ and $\text{Var}(Y_i) = 4$. Define the discrete-time random process $\{X_n, n \in \mathbb{N}\}$ as $X_n = Y_1 + Y_2 + \dots + Y_n$, for all $n \in \mathbb{N}$.

- Find $\mu_X(n)$ and $R_X(m, n)$, for all $n, m \in \mathbb{N}$.
- Is the process $\{X_n, n \in \mathbb{N}\}$ W.S.S?

2. For any $k \in \mathbb{Z}$, define the function $g_k(t)$ as $g_k(t) = \begin{cases} 1, & k < t \leq k+1, \\ 0, & \text{otherwise.} \end{cases}$

Now, consider the continuous-parameter random process $\{X_t, t \in \mathbb{R}\}$ defined as

$$X_t = \sum_{k=-\infty}^{+\infty} A_k g_k(t),$$

where $A_1, A_2, \dots$ are i.i.d. random variables with $\mathbb{E}[A_k] = 1$ and $\text{Var}(A_k) = 1$.

- Find $\mu_X(t)$, $R_X(s, t)$, and $C_X(s, t)$ for all $s, t \in \mathbb{R}$.
- Is the process $\{X_t, t \in \mathbb{R}\}$ W.S.S?

3. Let $\{X_t, t \in \mathbb{R}\}$ be a W.S.S process with mean $\mu_X = 1$ and $R_X(\tau) = \begin{cases} 4 - |\tau|, & -3 \leq \tau \leq 3, \\ 1, & \text{otherwise.} \end{cases}$

- Find $\mathbb{E}[X_t^2]$.
- Find $\mathbb{E}[(X_{10} + X_{12})^2]$.
- Find $\mathbb{E}[(X_t + X_{t+1.5} + X_{t+3})^2]$.

4. A biased random walk process is defined as $\sum_{i=0}^n U_i$, $n \geq 0$ where $\{U_i, i \geq 0\}$ is an i.i.d

Bernoulli process with PMF u_i being $P_{U_i}(k) = \begin{cases} 1/4, & k = -1, \\ 3/4, & k = 1, \\ 0, & \text{else} \end{cases}$.

Find $\mu_X[n]$ and $\text{Var}_X[n]$.

5. A random process $\{X_n : n \in \mathbb{N}\}$ is known to be W.S.S with $\mu_X[10] = 10$ and $\text{Var}_X[10] = 1$. Determine $\mu_X[100]$ and $\text{Var}_X[100]$.

6. Consider an IID Bernoulli process $\{X_n : n \in \mathbb{N}\}$ where $X_i \sim \text{unif}\{0, 1\}$. Let $Y_n = (-1)^n X_n$, $n \geq \mathbb{N}$. Is $\{Y_n : n \in \mathbb{N}\}$ an IID process? Justify.
7. Consider the continuous parameter process $X_t : t \in \mathbb{R}$, $X_t = A \cos(2\pi t)$, $t \in \mathbb{R}$ where A is a random variable $\sim \mathcal{N}(0, 1)$. Find $\mu_X(t)$ and $C_X(s, t)$.
8. Consider a random process $\{X_n : n \in \mathbb{Z}\}$ defined as $X_n = a_0 U_n + a_1 U_{n-1}$, where a_0, a_1 are constants and $\{U_n : n \in \mathbb{Z}\}$ is an IID process with mean $\mu_U[n] = 0 \forall n \in \mathbb{Z}$. Is $\{X_n : n \in \mathbb{Z}\}$ W.S.S? If yes, find $\mu_X[n]$ and $R_X[\Delta]$.
9. A random process $\{X_n : n \in \mathbb{Z}\}$ consists of independent random variables with

$$X_n \sim \begin{cases} \mathcal{N}(0, 1), & \text{for } n \text{ even,} \\ \text{unif}(-\sqrt{3}, \sqrt{3}), & \text{for } n \text{ odd.} \end{cases}$$

Is this process W.S.S? Is it stationary?

10. The random process $\{X_n : n \in \mathbb{N}\}$ and $\{Y_n : n \in \mathbb{N}\}$ are both W.S.S. In addition $X_n \perp\!\!\!\perp Y_m \forall n, m \in \mathbb{N}$.
 - Is $\{Z_n = X_n + Y_n : n \in \mathbb{N}\}$ W.S.S?
 - Is $\{W_n = X_n Y_n : n \in \mathbb{N}\}$ W.S.S?
11. Let $\{X_t : t \in \mathbb{R}\}$ be a Gaussian process where $X_t \sim \mathcal{N}(t, 1 + 2t^2)$ and $R_X(t_1, t_2) = 1 + 2t_1 t_2$, $\forall t_1, t_2 \in \mathbb{R}$. Find $\mathbb{P}\{2X_1 + X_2 < 3\}$.
12. Let $\{S_n : n \in \mathbb{N}\}$ be a Poisson process with rate λ . Here S_1 is the first arrival time. Let $\{N_t, t \in \mathbb{R}^+\}$ be the associated counting process. Show that, given $N_t = 1$, $S_1 \sim \text{unif}(0, t]$, i.e., prove that

$$\mathbb{P}(X_t \leq x | N_t = 1) = \frac{x}{t}, \text{ for } 0 \leq x \leq t.$$

Hint: Use the fact that $N_t \sim \text{Pos}(\lambda t)$.